Early Discoveries

- **Joseph Priestly**: Concluded that air is necessary for the growth of a plant. He discovered the fact that plants restore oxygen in the air, fouled because of burning and respiration of living organisms.
- **Jan Ingenhousz**: Concluded that sunlight is essential for plant processes that purify the air. He also discovered that the green parts of plants release oxygen.
- **Julius Von Sachs**: Discovered that the green substance in plants (chlorophyll) is located in special bodies (chloroplast). Green parts of plants are where glucose is made, which is stored as starch.
- **T.W. Engelmann**: Discovered the first action spectrum of photosynthesis using green alga *Cladophora*. It roughly resembles the absorption spectra of chlorophylls *a* and *b*.
- **Cornelius van Niel**: Demonstrated that photosynthesis is a light-dependent reaction in which hydrogen from an oxidisable compound reduces carbon dioxide to carbohydrates.

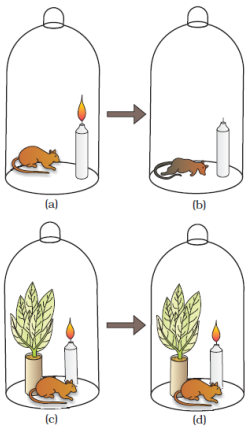


Figure Priestley's experiment

$$2H_2A + CO_2 \xrightarrow{Light} 2A + CH_2O + H_2O$$

- He worked on green and purple bacteria.
- When he worked on sulphur bacteria, he found that sulphur is the oxidation product in them, and not oxygen, since H_2S is the hydrogen donor there. Hence, he concluded that oxygen comes from H_2O, and not from CO_2.

Extra information

- **Ruben** and **Kamen** in 1941 proved that oxygen liberated during photosynthesis comes from water while working on Chlorella (unicellular green alga)
- When normal H_2O and radioactive CO_2 (i.e., CO_2^{18}) were used, normal O_2 evolved.

$$nCO_2^{18} + nH_2O \xrightarrow{light,chlorophyll} (CH_2O^{18})n + nO_2 \uparrow (normal) + H_2O$$

- When normal CO_2 and radioactive H_2O (i.e., H_2O^{18}) were used, radioactive O_2 (i.e., O_2^{18}) is evolved

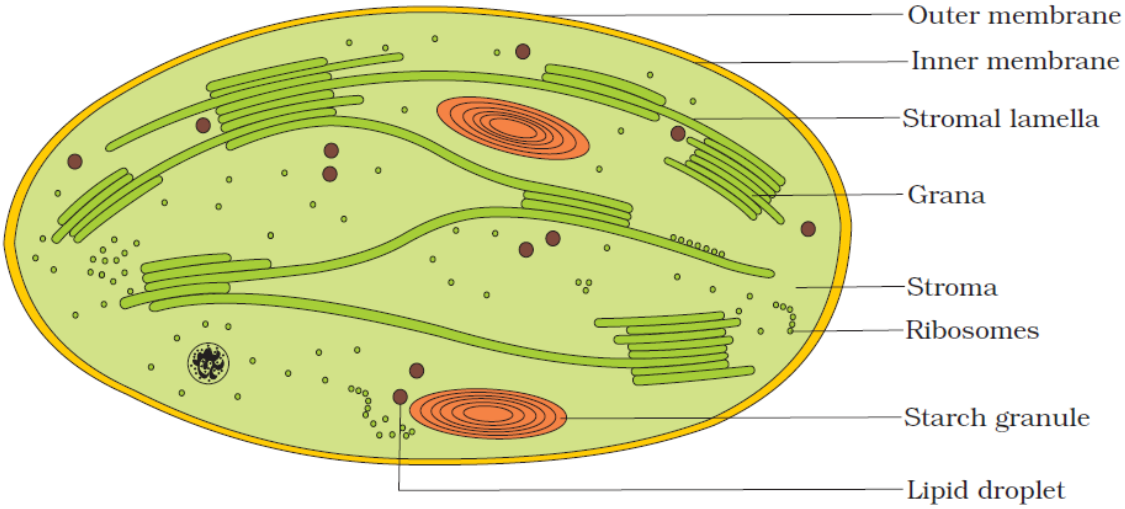
$$nCO_2 + nH_2O^{18} \xrightarrow{light,chlorophyll} (CH_2O)n + nO_2^{18} \uparrow (radioactive) + nH_2O$$

- Finally, the correct equation for photosynthesis was discovered.

$$6CO_2 + 12H_2O \xrightarrow{Light,chlorophyll} C_6H_{12}O_6 + 6H_2O + 6O_2$$

Site of Photosynthesis – Chloroplast

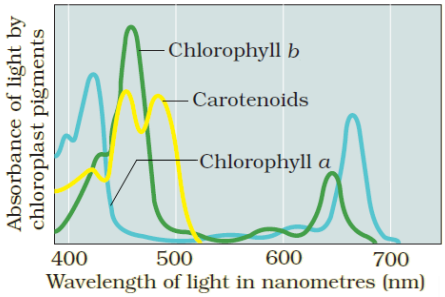
- Mesophyll cells in green leaves have large number of **chloroplasts** (Photosynthetic apparatus), which are the site of photosynthesis.
- Chloroplasts are aligned along the walls of mesophyll cells so as to get optimum light.
- Chloroplast consists of the grana and the stroma lamellae (forming a membranous system) and the fluid stroma.
- Membrane system of chloroplast – traps light, and synthesises ATP and NADPH (site of light-dependent reaction of photosynthesis)
- Stroma – CO_2 is incorporated into the plant by enzymatic reactions, leading to the synthesis of sugar (site of light-independent reaction of photosynthesis)



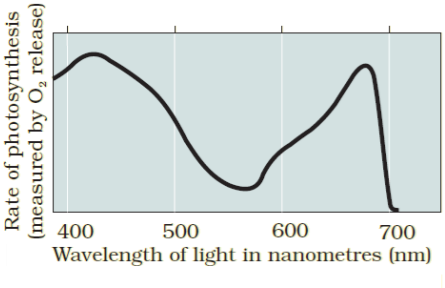
Diagrammatic representation of an electron micrograph of a section of chloroplast

Pigments Involved in Photosynthesis

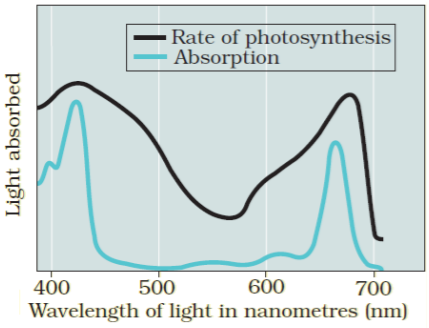
- An **absorption spectrum** is the graph plotted against the fraction of light absorbed by the pigment.
- An **action spectrum** (obtained from spectrophotometer) is the rate of a physiological activity plotted against the wavelength of light.
- The similarity of the action spectrum of photosynthesis and the absorption spectrum of chlorophyll tells us that chlorophylls are the most important pigments in the process.



(a) Graph showing the absorption spectrum of chlorophyll *a*, *b* and the carotenoids



(b) Graph showing action spectrum of photosynthesis



(c) Graph showing action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll *a*

- 4 types of pigments may be present in leaves:
  - Chlorophyll *a* ($C_{55}H_{72}O_5N_4Mg$) (blue-green)
  - Chlorophyll *b* ($C_{55}H_{70}O_6N_4Mg$) (yellow-green)
  - Xanthophylls ($C_{40}H_{56}O_2$) (yellow)
  - Carotenoids ($C_{40}H_{56}$) (yellow-orange) – Contains carotenes and xanthophylls

NOTE - These pigments are separated through paper chromatography as chromatogram

- Chlorophyll *a* is the main pigment in photosynthesis.
- In VIBGYOR spectra, chlorophyll *a* shows maximum absorption, and hence, the rate of photosynthesis is the highest at the blue and red regions.
- **Accessory pigments:** Chlorophyll *b*, xanthophylls and carotenoids

Significance of accessory pigments

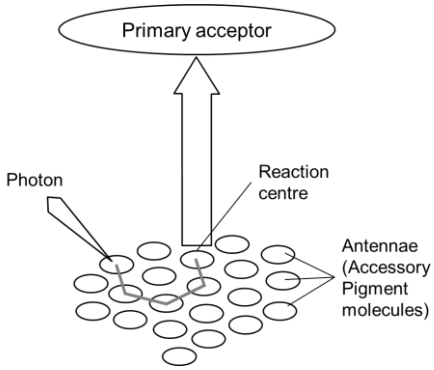
- Absorb a wider range of light, and transfer the energy to chlorophyll *a*
- Protect chlorophyll *a* from photo-oxidation

Light Reaction (Photochemical Phase)

- This phase directly depends on light. The pigments absorb light energy and produce ATP.
- Light reaction takes place in thylakoid membrane
- Includes:
  - Light absorption
  - Water splitting (Photolysis)
  - Oxygen release
  - Formation of ATP and NADPH, which is then used in the biosynthetic phase

- Pigment molecules bound to the proteins form **LHC** (light harvesting complexes). LHC are located within two photosystems – PSI and PSII
- Each photosystem has two parts:
  - Reaction centre – consisting of chlorophyll *a* molecule
  - Antennae – consisting of accessory pigments, which increase the efficiency of photosynthesis by absorbing different wavelengths of light

- Reaction centre is different in both photosystems:
  - PSI – P700; since chlorophyll *a* has absorption peak at 700 nm here
  - PSII – P680; since chlorophyll *a* has absorption peak at 680 nm here.



The light harvesting complex

Photo-Phosphorylation

- The process of formation of ATP in chloroplast in the presence of sunlight
- Photo-phosphorylation is of two types:
  - Non-cyclic photo-phosphorylation
  - Cyclic photo-phosphorylation

Non-Cyclic Photo-Phosphorylation (Z scheme)

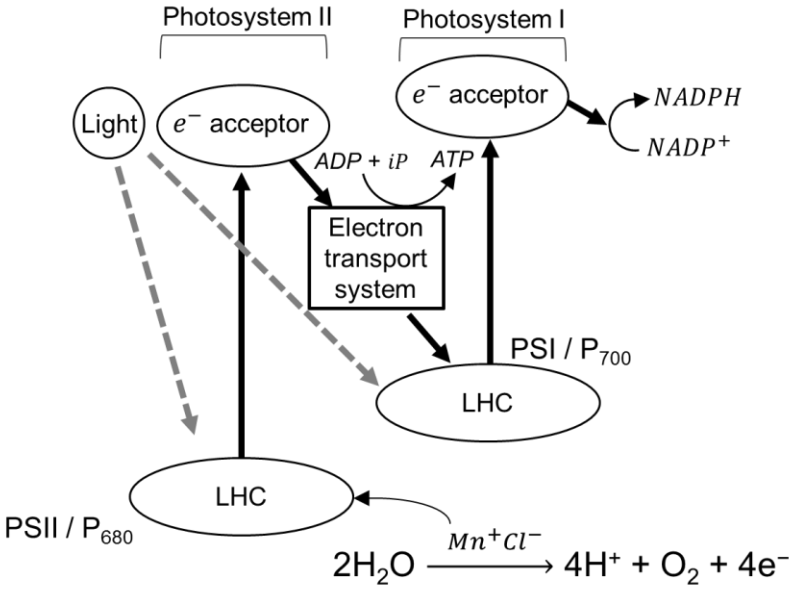


Figure: Z scheme of light reaction

- PSII absorbs 680 nm wavelength of red light, causing electrons to become excited and jump into an orbit farther from the nucleus.
- These electrons are then accepted by an electron acceptor (uphill e^- transport), which sends them to an electron transport system.
- Electron transport system transfers the electrons to PSI (downhill e^- transport).

- Electrons in PSI are simultaneously excited on receiving a wavelength of 700 nm. These electrons are again transferred to another electron acceptor having a greater redox potential (uphill e^- transport).
- From the electron acceptor, electrons are transferred to the molecule of $NADP^+$ (downhill e^- transport).
- Addition of these electrons reduces the $NADP^+$ to NADPH+ H^+ by NADP reductase enzyme.
- During this process, electrons move downhill, i.e., towards the systems at greater redox potential.
- The flow of electrons assumes the shape of the letter 'Z' when all carriers are placed according to their redox potential. Hence, the process is called **Z scheme.**
- Electrons lost by PSII do not come back to it.
- Synthesis of ATP in presence of light in chloroplast during non-cyclic transport of electrons is called **non-cyclic photophosphorylation**

Cyclic Photo-Phosphorylation

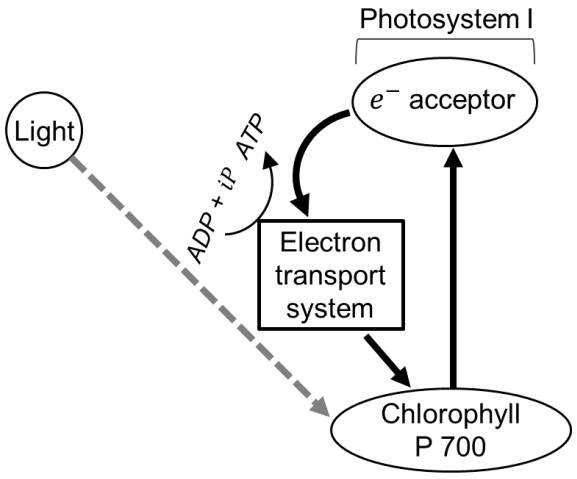


Figure: Cyclic photophosphorylation

- In this scheme, only PSI is functional. Hence, the electrons are circulated within the photosystem.
- This results in a cyclic flow of electrons.
- This scheme could possibly be occurring in stroma lamellae because it lacks both PSII and NADP reductase enzyme.
- This cyclic flow results only in the synthesis of ATP, and not of NADPH + H^+.
- Synthesis of ATP in chloroplast in presence of light during cyclic transfer of electron is called **cyclic photo-phosphorylation**.

Splitting of Water / Photolysis / Photocatalytic splitting of water

- The electrons being transferred in the photo-phosphorylation reactions are generated by the splitting of water.
- Water splitting complex is associated with PSII.
- Manganese, chlorine, etc., play an important role.
- The light-dependent splitting of water is called **photolysis**.

$$2H_2O \xrightarrow{Mn,Cl} 4H^+ + O_2 + 4e^-$$

- Electrons formed are used for replacing the electrons lost from P680.
- Oxygen is liberated as a byproduct of photosynthesis.
- Protons are used for the formation of reducing power NADP to $NADPH^+$.

NOTE - Robert Hill (1939), discovered that isolated chloroplasts when illuminated were capable of liberating oxygen in the presence of an oxidizing agent (electron acceptor). This has been called the **Hill reaction.** Since naturally the photolysis of water results in the oxidation of water molecules. the reaction is known as **Hill reacton**.

Chemiosmotic Hypothesis

- Chemiosmotic hypothesis postulated by **Peter Mitchell** in 1961 which explains how ATP is synthesised in chloroplast.
- Chemiosmosis requires a membrane, a proton pump, a proton gradient and ATPase.
- Membrane: Thylakoid membrane (impermeable to H^+); protons accumulate towards the inside of the membrane; in the lumen
- Proton gradient develops because of the following reasons:
  - Water splitting reaction takes place on the inner side of the membrane. So H^+ produced in the reaction accumulates within the lumen.
  - During the photo-phosphorylation steps when the electron-carrier molecule passes on its electrons to the electron-carriers on the inner side of the membrane, protons are released into the inner side or the lumen of the membrane.
  - NADP reductase enzyme is located on the stroma side of the membrane. Protons are removed from the stroma and provided to the lumen because protons are necessary for the reduction of $NADP^+$ to NADPH + H^+, along with the electrons that come from the electron- accepter located in PSI.

- As a result, the protons in the stroma decrease in number, while the protons in the lumen, increase in number, thus creating a proton gradient across the thylakoid membrane.
- The proton gradient is broken down due to the movement of H^+ from the lumen towards the stroma through the transmembrane channel of the F_0 of ATPase.
- ATPase: This enzyme has two parts
  - CF_0 – embedded in the membrane and carry out facilitated diffusion of H^+
  - CF_1 – protrudes towards the stroma

- Conformational changes are induced in the F_1 particle of the ATPase. This makes ATPase synthesise ATP molecules.
- Since formation of ATP is due to the diffusion of protons by diffusion force similar to osmosis, this process is referred to as **chemiosmosis**

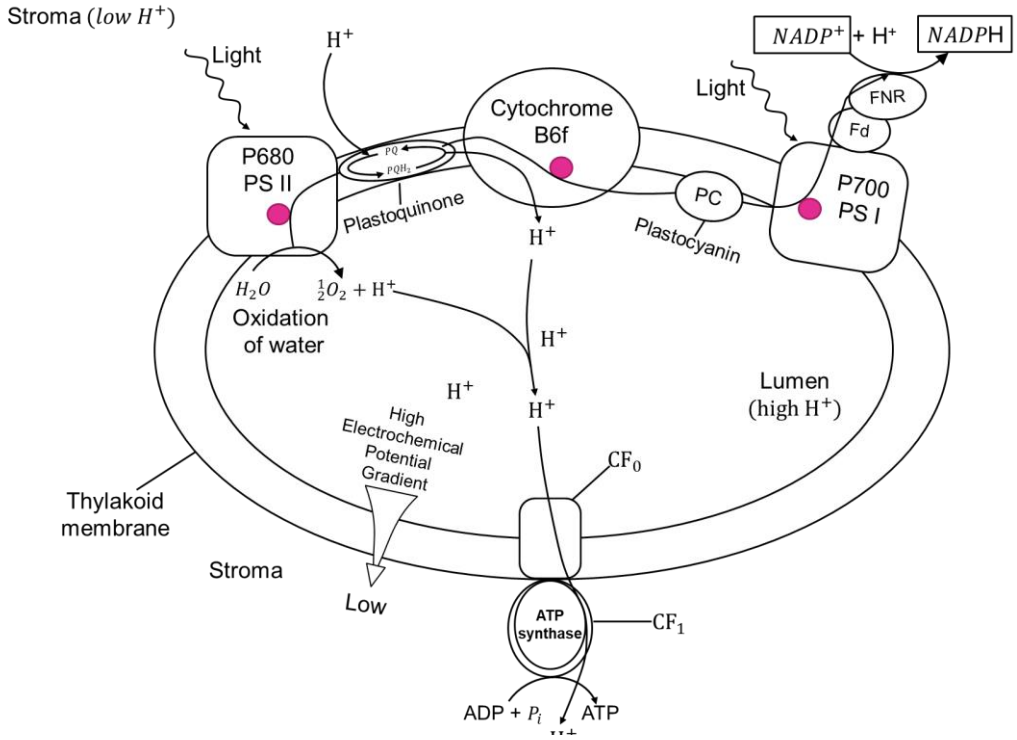


Figure: ATP synthesis through chemiosmosis

NOTE – Evidence suggested that single ATP is formed for every 3 H^+ passing outwards through F_0F_1 particles

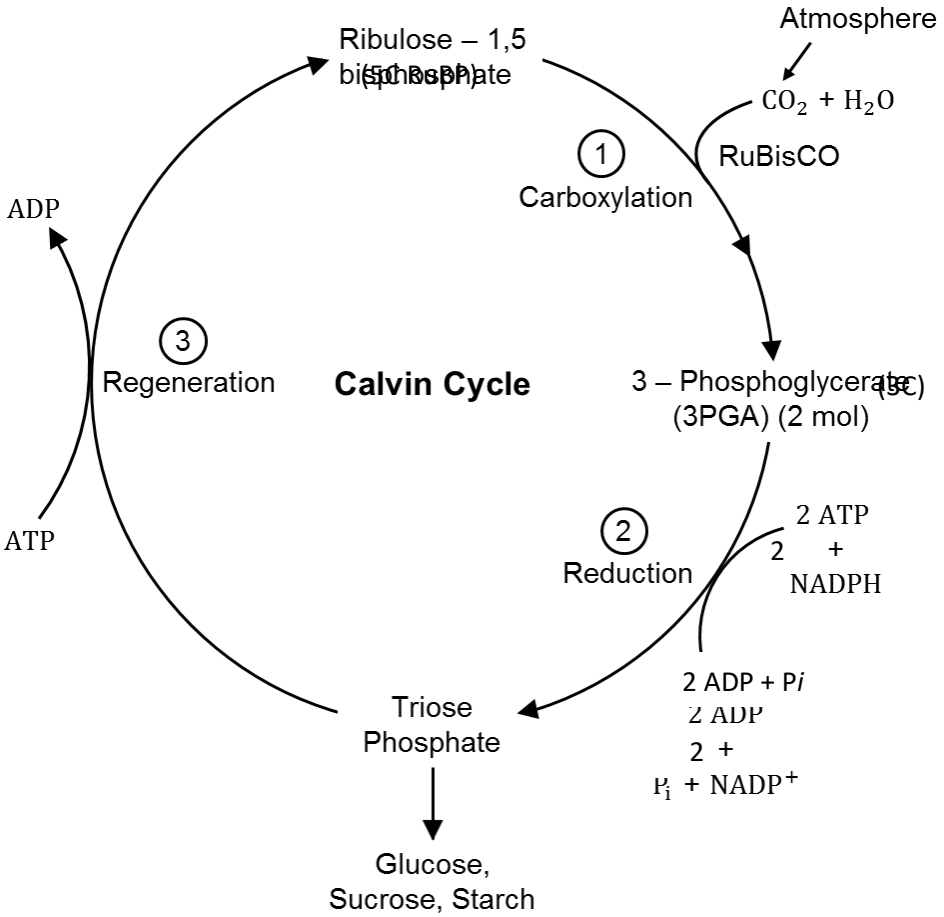
Dark Reaction/Blackman's reaction (Biosynthetic Phase)

- Products of light reaction: ATP, NADPH, CO_2 and H_2O
- Next stage is the biosynthetic phase. In this, ATP and NADPH are used for synthesising the food.
  - CO_2 is fixed to form CO_2 fixation product.
  - This stage is also called the dark phase as it is independent of light, but deals with the products of light reaction.

- The reaction involved in the biosynthetic phase takes place in the stroma of chloroplasts.
- In some plants, the first product of CO_2 fixation is a 3-carbon compound called 3-phosphoglyceric acid (PGA). These plants are said to adopt the C_3 pathway.
- In other plants, the first CO_2 fixation product is a 4-carbon compound called oxaloacetic acid. These plants are said to adopt the C_4 pathway.

Calvin Cycle (C_3 Cycle)

- The path of carbon in the dark reaction was traced by **Melvin Calvin** using radioactive carbon (^{14}C).
- Calvin pathway occurs in all photosynthetic plants, irrespective of whether they have C_3 or C_4 pathway.
- The primary acceptor of CO_2 was found to be a 5-carbon ketose sugar called **Ribulose bisphosphate** (RuBP). This was surprising as it was believed that for the formation of a 3-carbon product, CO_2 would have to be accepted by a 2-carbon compound.
- RuBP is used in a cyclic manner (regenerated) and a sugar is synthesised.
- Example – *Helianthus* (sunflower)
- 3 stages of Calvin cycle: Carboxylation, Reduction and Regeneration



Carboxylation

- Ribulose 1, 5-bisphosphate combines with CO_2, and fixes it to a stable organic intermediate (3-phosphoglycerate – 2 molecules).
- Reaction catalysed by the enzyme RuBisCO (RuBP Carboxylase-Oxygenase)
- 3 PGA is the first stable product of this cycle.

Reduction

- Here, 2 molecules each of ATP and NADPH are required for fixing 1 molecule of CO_2.
- This stage contains a series of reactions.
- Glucose is formed as a result of this series of reactions.

Regeneration

- RuBP regenerates to enable the cycle to continue uninterrupted.
- 1 ATP molecule is required.

- For the formation of one molecule of glucose, six molecules of CO_2 need to be fixed; hence, six cycles are required.

ATP required:

- For fixing 1 molecule of CO_2 – 3 (2 for reduction and 1 for regeneration)
- For fixing 6 molecules of CO_2 – $3 \times 6 = 18$ ATP

NADPH required:

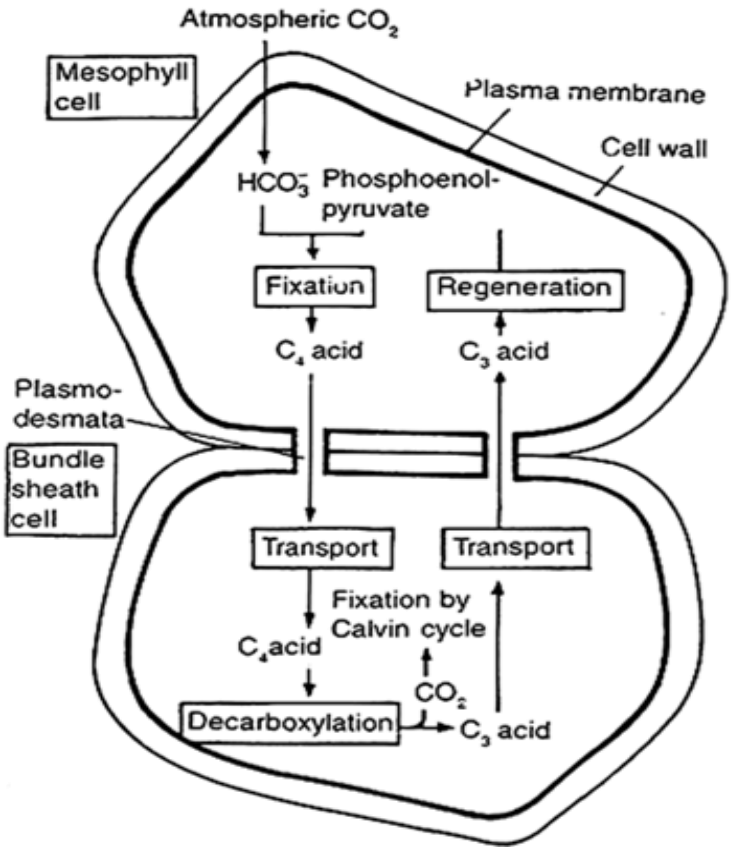
- For fixing 1 molecule of CO_2 – 2 (for reduction)
- For fixing 6 molecules of CO_2 – $2 \times 6 = 12$ NADPH

- Thus, the synthesis of 1 molecule of glucose requires 18 ATP and 12 NADPH.

C_4 Pathway / Hatch and Slack Pathway

- Occurs in plants like maize, sugarcane – plants adapted to dry tropical regions
- The leaves of C_4 plants have **Kranz anatomy**. They have special large cells around their vascular bundles called bundle sheath cells. These cells form several layers around the vascular bundles and have thick walls which is impermeable to gaseous exchange, and lack intercellular spaces and contains large chloroplasts.
- First CO_2 fixation stable product is a 4-carbon compound called **oxaloacetic acid** (OAA), but C_3 cycle is used as the main biosynthetic phase.
  - C_4 cycle Occur in the Mesophyll cells
  - C_3 cycle Occur in the Bundle Sheath Cells

- C_4 plants can tolerate high temperature and high light intensity, show greater productivity of biomass, and lack photorespiration.
- Primary CO_2 acceptor: Phosphoenol pyruvate (PEP) – a 3-carbon molecule
- PEPcase (PEP Carboxylase) fixes CO_2 in the mesophyll cells. It forms the 4-carbon compound oxaloacetic acid (OAA), and then other 4-carbon compounds like malic acid and aspartic acid.
- These compounds are transported to the bundle sheath cells. There, C_4 acid breaks down to form C_3 acid and CO_2 (released only in the bundle sheath cells, and enters the C_3 cycle).
- C_3 acid, so formed, is again transported to the mesophyll cells and regenerated back into PEP.
- C_3 cycle cannot directly occur in the mesophyll cells of C_4 plants because of the lack of the enzyme RuBisCO in these cells
- RuBisCO is found in abundance in the bundle sheath cells of C_4 plants. They lack PEPcase.



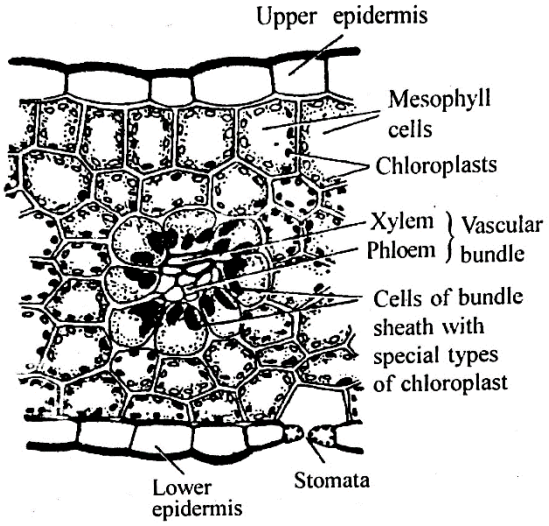
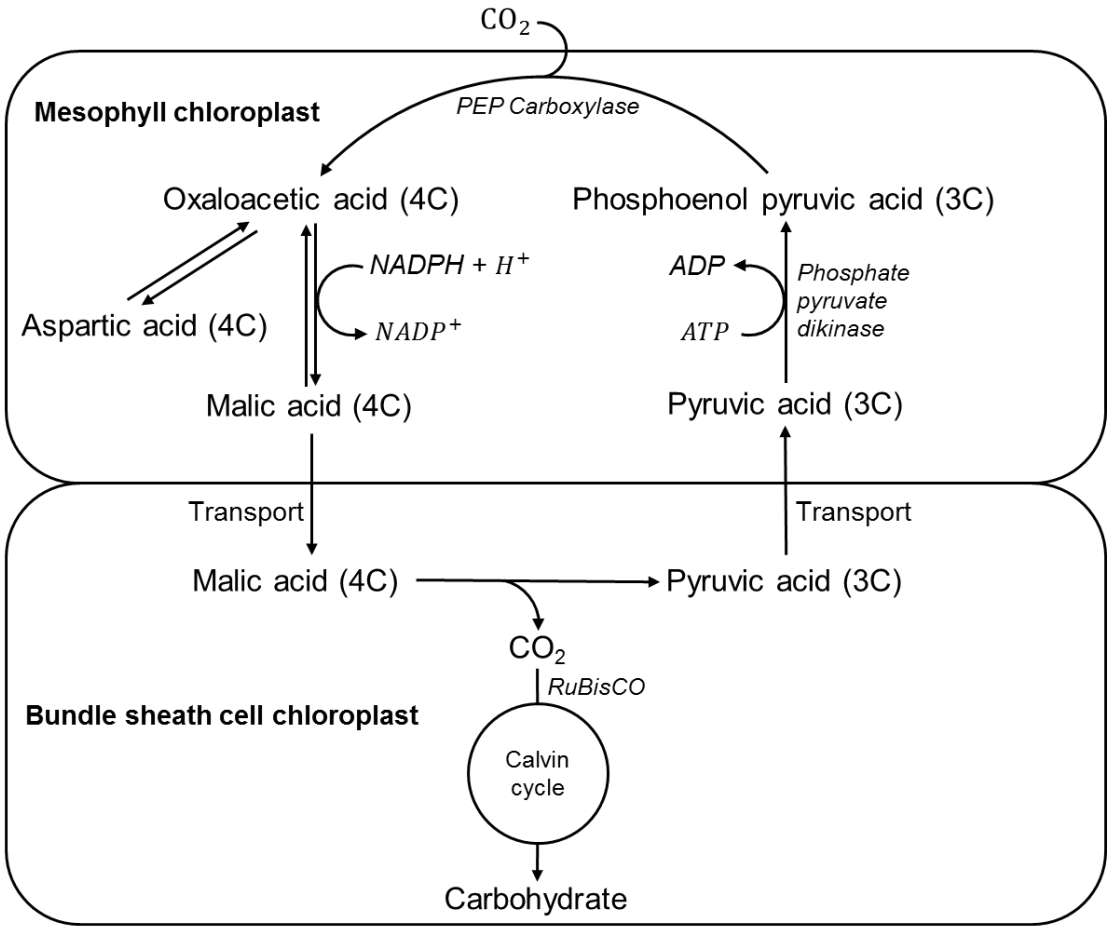


Figure: Cross section of leaf showing Kranz anatomy

Detailed illustration of C4 Cycle



Photorespiration (C2 Pathway)

- Photorespiration is the respiration that occurs in green cells in the presence of light resulting in the evolution of CO_2.
- It was observed by **Decker** in 1955
- It is a process in which there is no formation of ATP or NADPH or sugar but there is utilization of ATP with release of CO_2. Hence photorespiration in C3 plants is considered a wasteful process.
- To understand this, let us first recall the first step of Calvin pathway.

  $RuBP + CO_2 \xrightarrow{RuBisCO} 2 \text{ X } 3PGA$
- This reaction is catalyzed by an enzyme RuBisCO. The active site of RuBisCO can bind to both O_2 and CO_2, though it has a greater affinity for CO_2.
- In C_3 plants, some of the O_2 binds to RuBisCO to form phosphoglycerate and phosphoglycolate (2C) (instead of PGA). There is no production of sugar or ATP. This mechanism in C_3 plants results in utilization of ATP with the release of CO_2.
- Since 2 carbon compound phosphoglycolate is produced after joining of O_2 with RuBP is also referred as **C2 pathway**
- Photorespiration is responsible for the difference between C_3 and C_4 plants.

Do C_4 plants have better yield than C_3 plants?

- In C_4 plants, photorespiration does not occur since they have mechanism to increase the CO_2 concentration at enzyme site. C_4 acids are broken in the bundle sheath cells to release CO_2, thus increasing the intracellular concentration of CO_2.
  This is the reason why C_4 plants are better yielding as compared to C_3 plants.

RuBisCO - A unique enzyme!

RuBisCO or ribulose bisphosphate carboxylase oxygenase is the most abundant enzyme in the world!

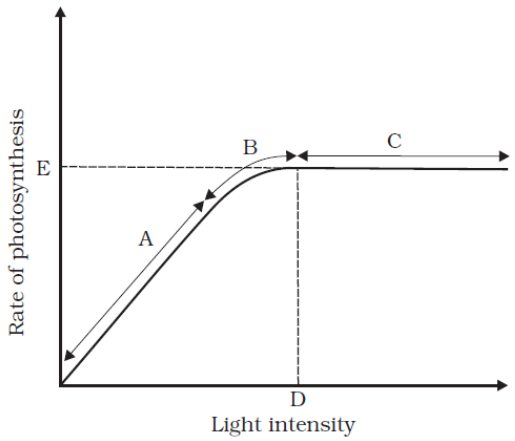
As the name suggests, it can bind to both carbon dioxide as well as oxygen. The binding to the enzyme is determined by the relative concentration of oxygen and carbon dioxide.

Factors Affecting Rate of Photosynthesis – Blackman law of limiting factor

- Photosynthesis is influenced by internal (plant) factors and external factors.
- **Limiting Factor –** If a chemical process is affected by more than one factor, then its rate will be determined by factor which is nearest to its minimal value (factor which directly affects the process if its quantity is changed).
- This law of limiting factor was given by **Blackman** (1905).

Light

- Incident light $\propto$ CO_2 fixation rate; but at higher light intensities, the rate does not increase further as other factors become limiting
- Light is rarely a limiting factor (with exception of the shade plants or plants of dense forest) because light saturation occurs at 10% of the full sunlight.
- Beyond a point, if incident light is increased, then it leads to decrease in photosynthesis due to breakdown of chlorophyll and finally death of cell. This phenomenon is called **solarization.**



Graph of light intensity on the rate of photosynthesis

Where

'A' and 'B' represent that the light is limiting factor in this region

'C' represents that rate of photosynthesis is not increases further by increasing light intensity because some other factor becomes limiting.

'D' represents the intensity of light at which some other factor became limiting

CO_2 Concentration

- Major limiting factor
- Usually low in atmosphere (0.03 – 0.04%)
- Up to 0.05% – increases rate of CO_2 fixation
- 0.05% – damaging effect
- Though both C_3 and C_4 show increase in rate of photosynthesis at high light intensity accompanied by high CO_2 concentration, the saturation point for C_3 is obtained at higher concentrations as compared to C_4. Therefore, CO_2 concentration is more of a limiting factor for C_3 plants.
- Increased CO_2 concentration is beneficial for greenhouse crops such as tomatoes and bell pepper (capsicum).

Temperature

- Dark reactions are more sensitive to an increase in temperature.
- C_4 plants respond more to an increase in temperature $(30 - 45^0C)$ as compared to C_3 plants $(10 - 25^0C)$.
- Adaptations according to habitat also effect temperature optima. Tropical plants have higher temperature optima compared to temperate.

Water

- Water stress causes stomata to close and hence, less CO_2 is available.
- Water stress causes the leaves to wilt (bending due to loss of water), thereby reducing their surface area and metabolic activity as well.